

Photosynthesis: the light lab

Teacher guide and worked answers

Describe photosynthesis and investigate how light intensity affects a measured proxy for its rate.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Spring | C36

Describe photosynthesis (core practical; data-logging).

Directly develops C36 and Pearson 6.2,6.3 and core practical 6.5. Supplied data support planning/analysis if a practical cannot run; they do not count as pupils personally completing a practical. Inverse-square calculations are not required.

Prior learning

Plants have chloroplasts containing chlorophyll.

A fair comparison changes one intended variable.

Rate means an amount or change per unit time.

Success evidence

State the reactants, products and energy source for photosynthesis.

Explain a fair light investigation with a measured outcome and controls.

Calculate a mean bubble rate and explain its limitation as a proxy.

Equipment

Printed shared evidence and one selected response route; pen, ruler and calculator.

Projector or offline HTML/PowerPoint. Optional word processor or spreadsheet.

Optional supervised demonstration: school-approved aquatic plant, transparent vessel on tray, same dilute sodium hydrogencarbonate solution, cool LED lamp, ruler, timer and thermometer.

Optional light sensor/data logger with lux display; never rely on distance alone as a direct intensity measurement.

Preparation

Print the shared evidence page, one chosen response route and exit page. Routes are alternatives, not a workload ladder.

Read the teacher answers before teaching. All numerical sample data in this lesson are constructed for learning.

The paper/data route is complete. If demonstrating, technician/teacher prepares an approved pondweed shoot with a fresh cut under water and checks bubbling before the lesson. Do not acquire or release invasive species.

Keep the same shoot, solution volume/concentration, counting interval and background light. Let the shoot adjust for at least 2 minutes after a light change; monitor temperature. Record actual results separately from constructed values.

For data logging, position the light sensor consistently beside the vessel facing the lamp; log one reading each second for 30 seconds at each position and record the mean. Lux is a practical light indicator, not a direct measurement of photosynthetic photon flux.

Practical care

Use the fully supplied paper route if apparatus is not prepared. No flames, gas tests or hot-lamp practical is required.

Teacher handles cutting and fragile glass. Keep electrical equipment away from water; use a stable tray and school-approved lamp setup.

Do not taste solution or plant material. Wash hands after handling and dispose of plant material according to school procedures; never release it outdoors.

Ready to teach alternative

Analyse the supplied constructed table and apparatus diagram. Pupils make the same predictions, controls and evidence-based conclusions without using water, plants or electrical equipment.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Can light change what a plant makes?
2	Retrieval	2-5	Materials, energy and measurement
3	I do	5-9	Read the reaction accurately
4	I do	9-12	I plan the measurement before moving the lamp
5	I do	12-15	Work a rate and a mean
6	We do	15-18	Investigation team: predict, then calculate
7	We do	18-20	Method repair challenge
8	We do	20-22	Which conclusion earns its claim?
9	Hinge	22-25	A valid scientific conclusion
10	You do	25-26	Prepare your investigation brief
11	You do	26-36	Use the evidence independently
12	Review	36-38	Audience question: what did you measure?
13	Review	38-39	A strong brief names its limitation
14	Exit	39-40	Keep matter, energy and evidence distinct
15	Exit	Within stage	Next useful step

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Can light change what a plant makes?

Ask pupils to predict the effect of moving the lamp closer. State that this is a planned or supervised comparison, not a race to produce the most bubbles.

2 Retrieval Materials, energy and measurement

R1 chloroplasts. R2 kept the same to improve fairness. R3 a 60-second interval. If pupils say plant food comes from soil, revisit glucose production in the next diagram.

3 I do Read the reaction accurately

Think aloud: "I list matter on the two sides and put energy above the arrow. Glucose is a product, not a soil nutrient. Plants also respire; this reaction is not their only process." Teach endothermic as takes in energy without requiring a balanced symbol equation.

4 I do I plan the measurement before moving the lamp

Model check: same shoot and solution; at least 2 minutes adjustment; one-minute counts repeated three times; temperature monitored and kept comparable. Use sensor consistently. A cooler lamp reduces a heating confound, but does not remove the need to check.

5 I do Work a rate and a mean

Model $12 \div 2 = 6$ bubbles/min. Then $(4+5+6) \div 3 = 5$. These are constructed examples. Explain differing bubble sizes and dissolved gas. A gas-volume measurement over time can improve the proxy, but it still needs controlled conditions.

6 We do Investigation team: predict, then calculate

Use supplied data or own demonstration data with labels kept distinct. Constructed means 5,10,17. Ask pupils whether the measured lux values support the closer-lamp prediction; do not introduce a proportional-distance rule.

7 We do Method repair challenge

No: differences in shoot length/leaf area/activity may change bubbling. Keep same shoot or matched replicates and change intended light level only, while monitoring temperature and other controls. Ask a pupil to justify what the repair isolates.

8 We do Which conclusion earns its claim?

It supports the observed count pattern in these conditions. It does not establish unlimited response, exact oxygen volume or plant growth. Ask for the missing measurement needed to claim growth.

Reveal: Mean count rises $5 \rightarrow 10 \rightarrow 17$ bubbles/min in this constructed range. Other limiting factors and proxy limits remain.

9 Hinge A valid scientific conclusion

B correct. Return to reaction if A, bubble limitation if C, controls if D.

A. Oxygen is made from light as a material. - Light supplies energy; carbon dioxide and water are the material reactants.

B. Bubble count can indicate a pattern, but bubble size limits precision. - Yes. Count is a proxy; equal counts need not mean equal oxygen volume.

C. Twice the bubbles always means exactly twice the oxygen volume. - Bubble sizes can differ and some oxygen dissolves.

D. Changing the shoot and light together isolates light. - The shoot change is another possible cause of a count difference.

10 You do Prepare your investigation brief

Provide shared data, selected route and calculator. A labelled diagram or oral explanation can show the same core understanding.

11 You do Use the evidence independently

Supported S1-S4; standard M1-M4; stretch T1-T4. Check rate units and control reasoning. Do not award an unsupported claim that light always remains limiting.

12 Review Audience question: what did you measure?

Pupil names the measured bubble count and explains its use as proxy; revise overclaim. Teacher/TA can be a quiet audience.

13 Review A strong brief names its limitation

Model: "More measured light was associated with higher mean bubble counts under matched conditions. Bubble size and dissolved gas limit how exactly counts show oxygen production."

14 Exit Keep matter, energy and evidence distinct

E1 carbon dioxide/water. E2=5 bubbles/min. E3 same shoot/solution/temp/timing; bubble size or gas dissolving. Use errors to identify next practical support.

15 Exit Next useful step

Shared exit minute; accept labelled arrows instead of prose.

Answers and assessment

Retrieval

R1 chloroplasts. R2 a factor kept the same to improve fairness. R3 each 60 seconds.

W1-W3

W1 means 5,10,17 bubbles/min at 300,600,1200 lux. W2 use the same shoot/matched replicates and change the intended light level only, controlling temperature and solution. W3 count rises with measured light in this range; no unlimited-growth or exact-volume claim is justified.

Hinge

A false: light is energy, not a material reactant. B true: count indicates a pattern with bubble-size limitations. C false: different bubble sizes/dissolved gas. D false: changing shoots introduces another variable.

S1 / M1

Carbon dioxide + water → glucose + oxygen, with light energy absorbed by chlorophyll. Photosynthesis takes in energy; glucose can contribute to biomass or respiration.

S2 / M2

$(4+5+6)/3=5$; $(9+10+11)/3=10$; $(16+17+18)/3=17$ bubbles/min. From 300 to 1200 lux the mean increases by 12 bubbles/min. This is the observed range, not a universal proportionality.

S3 / M3

Change measured light level using lamp position. Count bubbles for equal one-minute intervals; repeat three times after adjustment. Same shoot, solution volume/concentration, background light and

comparable temperature. Sensor orientation/location consistent; optional 30 second logging mean. Controls reduce alternative explanations.

S4 / M4

Higher measured light is associated with higher mean count here, for example 300 lux → 5 and 1200 lux → 17 bubbles/min. Counts do not equal exact gas volumes because sizes differ and gas may dissolve. Collect gas volume over a fixed time with appropriate apparatus to reduce the bubble-size limitation; maintain other controls.

T1

$(16+17+35)/3=68/3\approx22.7$ bubbles/min only if all counts were truly one minute. If the 35 count lasted two minutes, averaging raw counts as equal rates is invalid. The note is uncertain; check the record before recalculating.

T2

Retain and annotate the original record; verify timing or repeat matched measurements with a clear timer/observer. Do not silently delete an inconvenient value or assume the cause from its size alone.

T3

Carbon dioxide availability or temperature could limit the rate. Example: compare matched carbon dioxide availability at the same light and temperature with repeats. The plateau alone does not identify which factor is limiting.

T4

Lux measures light weighted to human visual sensitivity, not glucose production or exact photosynthetic photon flux. Bubble counts indicate gas release, with volume/dissolution limits; glucose itself is not measured.

Exit and assessment

E1 carbon dioxide and water. E2 $15\div3=5$ bubbles/min. E3 valid controlled factor plus bubble-size/dissolved-gas limit. L1 accepts a specific evidence-based revision or justified label. Secure core = correct reaction, fair variable logic and proxy-aware conclusion.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 10, We do 7, hinge 3, independent 11, review 3, exit 1. Zero-minute slides share their stage time.

Choose one response route. Accept accurate speech, pointing plus explanation, annotations or writing as evidence.

A quiet review still has an audience: teacher or TA reads the pupil's decision, asks one evidence question, and the pupil revises or confirms it.

Optional apparatus is a supervised demonstration/practical route only after local preparation. The complete supplied data route teaches planning and analysis; do not record it as pupils having personally completed a core practical.

Interactive We do uses the exact three constructed light/count conditions. Predict which condition gives the greatest mean, test the result, then check whether the method isolates light. Paper W1-W3 uses the same table. Any animated bubbles illustrate a constructed count, not a physiological simulation.

Access and responsive teaching

Supported

Same core scientific goal, with word bank, short prompts, read-aloud or pointing/dictation. Do not assign a route from diagnosis.

Standard

Use the supplied evidence to explain and justify an independent scientific decision.

Stretch

Optional challenge after secure core: evaluate a limitation and transfer the science. Higher-tier-only material, where present, is labelled explicitly.

Regulation

Preview the sequence. Offer solo or partner work, a pass and quiet response. Allow a brief reset and re-entry at the next numbered task; no compulsory disclosure, performance or speed competition.

Misconceptions to check

Light is a material reactant that becomes oxygen.

Light supplies energy; carbon dioxide and water are reactants, glucose and oxygen are products.

Check: Ask pupils to put light over the reaction arrow.

A bubble count is an exact oxygen volume.

Bubble sizes can differ and some gas dissolves; the count is an imperfect proxy.

Check: Compare one large bubble with two small ones.

Changing lamp distance changes only light.

It can also change heating; monitor temperature and use a cool source/appropriate control.

Check: Ask which measurement checks an unwanted temperature change.

More light always increases photosynthesis without limit.

At higher light another factor can limit the rate; the supplied range is not a universal rule.

Check: Ask whether three conditions justify an unlimited prediction.

Word	Meaning
photosynthesis	Using light energy to make glucose from carbon dioxide and water, releasing oxygen.
chlorophyll	A pigment that absorbs light in photosynthesis.
rate	How much happens in a stated time.
proxy	A measurement used to indicate another quantity, with limitations.
control variable	A factor kept the same to make a comparison fairer.
limiting factor	A factor in short supply that restricts the rate of a process.

Scientific references

Pearson Edexcel GCSE Biology 1BI0 specification

Used to check relevant Foundation curriculum content; this is teacher-authored practice, not an official assessment or complete unit. Checked 8 September 2026.

SAPS: Demonstrating oxygen evolution using pondweed

Checked pondweed bubble investigation and the light comparison. All distances, sensor readings and counts below are constructed. Checked 8 September 2026.

SAPS: Measuring the rate of photosynthesis

Checked oxygen-volume and bubble-count methods and their interpretation. Checked 8 September 2026.